
Functional Dependencies and Normalization for Relational Databases

Informal Design Guidelines for Relational Databases

- ◆ Relational database design: The grouping of attributes to form "good" relation schemas
- ◆ Two levels of relation schemas:
 - The logical "user view" level
 - The storage "base relation" level
- ◆ Design is concerned mainly with **base relations**
- ◆ Criteria for "good" base relations:
 - Discuss informal guidelines for good relational design
 - Discuss formal concepts of functional dependencies and normal forms 1NF 2NF 3NF BCNF

Semantics of the Relation Attributes

- ◆ Each tuple in a relation should represent one entity or relationship instance
 - Only foreign keys should be used to refer to other entities
 - Entity and relationship attributes should be kept apart as much as possible
 - Design a schema that can be explained easily relation by relation. The semantics of attributes should be easy to interpret.
 - Tip: do not mix attributes from multiple entities

Figure 14.1 Simplified version of the COMPANY relational database schema.

EMPLOYEE

ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUMBER
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p.k.

DEPARTMENT

DNAME	<u>DNUMBER</u>	DMGRSSN
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p.k.

DEPT_LOCATIONS

<u>DNUMBER</u>	<u>DLOCATION</u>
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p.k.

PROJECT

PNAME	<u>PNUMBER</u>	PLOCATION	DNUM
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p.k.

WORKS_ON

<u>SSN</u>	<u>PNUMBER</u>	HOURS
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p.k.

EMPLOYEE

ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUMBER
Smith,John B.	123456789	1965-01-09	731 Fondren,Houston,TX	5
Wong, Franklin T.	333445555	1955-12-08	638 Voss,Houston,TX	5
Zelaya,Alicia J.	999887777	1968-07-19	3321 Castle, Spring,TX	4
Wallace,Jennifer S.	987654321	1941-06-20	291 Berry,Bellaire,TX	4
Narayan,Remesh K.	666884444	1962-09-15	975 Fire Oak,Humble,TX	5
English,Joyce A.	453453453	1972-07-31	5631 Rice,Houston,TX	5
Jabbar,Ahmad V.	987987987	1969-03-29	980 Dallas,Houston,TX	4
Borg,James E.	888665555	1937-11-10	450 Stone,Houston,TX	1

DEPARTMENT

DNAME	<u>DNUMBER</u>	DMGRSSN
Research	5	333445555
Administration	4	987654321
Headquarters	1	888665555

DEPT_LOCATIONS

<u>DNUMBER</u>	<u>DLOCATION</u>
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

<u>SSN</u>	<u>PNUMBER</u>	HOURS
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0

PROJECT

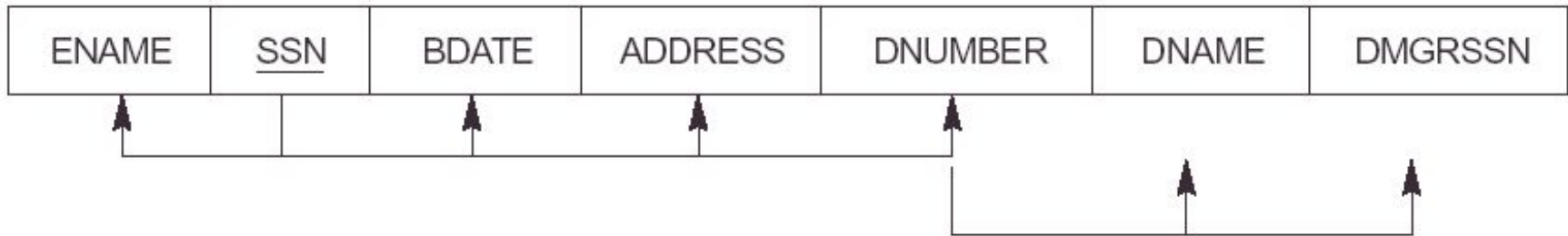
PNAME	<u>PNUMBER</u>	PLOCATION	DNUM
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

Redundant Information in Tuples and Update Anomalies

- ◆ Mixing attributes of multiple entities may cause problems
 - Information is stored redundantly wasting storage
 - Problems with update anomalies:
 - Insertion anomalies
 - Deletion anomalies
 - Modification anomalies

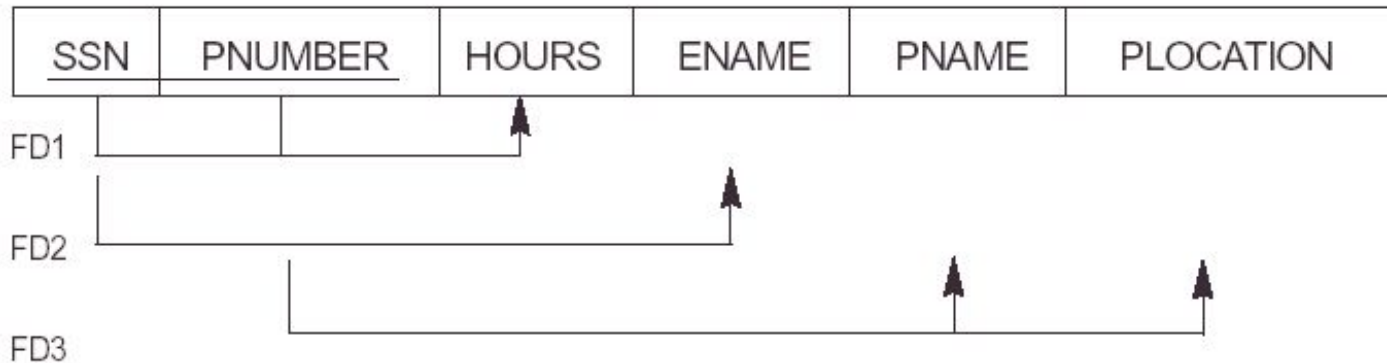
(a)

EMP_DEPT



(b)

EMP_PROJ



- (a) Is combined details of employee and department entities
- (b) Is combined details of employee and project entities

EMP_DEPT

ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUMBER	DNAME	DMGRSSN
Smith,John B.	123456789	1965-01-09	731 Fondren,Houston,TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss,Houston,TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring,TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry,Bellaire,TX	4	Administration	987654321
Narayan,Ramesh K.	666884444	1962-09-15	975 FireOak,Humble,TX	5	Research	333445555
English,Joyce A.	453453453	1972-07-31	5631 Rice,Houston,TX	5	Research	333445555
Jabbar,Ahmad V.	987987987	1969-03-29	980 Dallas,Houston,TX	4	Administration	987654321
Borg,James E.	888665555	1937-11-10	450 Stone,Houston,TX	1	Headquarters	888665555

EMP_PROJ

<u>SSN</u>	<u>PNUMBER</u>	HOURS	ENAME	PNAME	PLOCATION
123456789	1	32.5	Smith,John B.	ProductX	Bellaire
123456789	2	7.5	Smith,John B.	ProductY	Sugarland
666884444	3	40.0	Narayan,Ramesh K.	ProductZ	Houston
453453453	1	20.0	English,Joyce A.	ProductX	Bellaire
453453453	2	20.0	English,Joyce A.	ProductY	Sugarland
333445555	2	10.0	Wong, Franklin T.	ProductY	Sugarland
333445555	3	10.0	Wong, Franklin T.	ProductZ	Houston
333445555	10	10.0	Wong, Franklin T.	Computerization	Stafford

EXAMPLE OF AN UPDATE ANOMALY

Consider the relation:

EMP_PROJ (Emp#, Proj#, Ename, Pname, No_hours)

- **Update Anomaly**

- Changing the name of project number 1 from “ProductY” to “Customer-Accounting” may cause this update to be made for all 3 employees working on project 1

- **Insert Anomaly**

- Cannot insert a project unless an employee is assigned to .
- Inversely- Cannot insert an employee unless he/she is assigned to a project.

EXAMPLE OF AN UPDATE ANOMALY (2)

- **Delete Anomaly**

- When a project is deleted, it will result in deleting all the employees who work on that project. Alternately, if an employee is the sole employee on a project, deleting that employee would result in deleting the corresponding project.

- ◆ Design a schema that does not suffer from the insertion, deletion and update anomalies. If there are any present, then note them so that applications can be made to take them into account

Null Values in Tuples

- ◆ Relations should be designed such that their tuples will have as few NULL values as possible
 - Attributes that are NULL frequently could be placed in separate relations (with the primary key)
 - Reasons for nulls:
 - a. attribute not applicable or invalid
 - b. attribute value unknown (may exist)
 - c. value known to exist, but unavailable

Spurious Tuples

- ◆ Bad designs for a relational database may result in erroneous results for certain JOIN operations
- ◆ The "lossless join" property is used to guarantee meaningful results for join operations
- ◆ The relations should be designed to satisfy the lossless join condition. No spurious tuples should be generated by doing a natural-join of any relations

Informal Guidelines

Guideline 1:

- Informally, each tuple in a relation should represent one entity or relationship instance. (Applies to individual relations and their attributes).

Guideline 2:

- Design a schema that does not suffer from the insertion, deletion and update anomalies.
- If there are any present, then note them so that applications can be made to take them into account

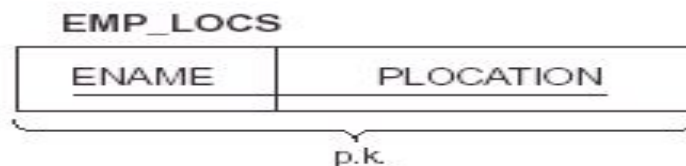
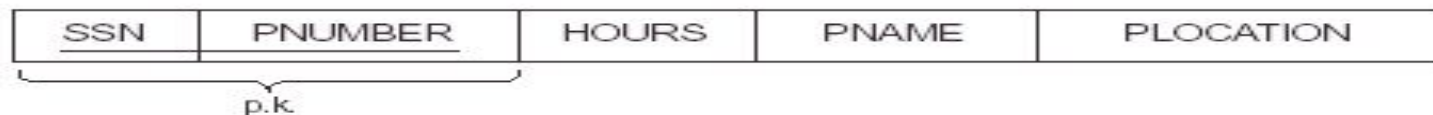
Guideline 3:

- Relations should be designed such that their tuples will have as few NULL values as possible
- Attributes that are NULL frequently could be placed in separate relations (with the primary key)

Guideline 4:

- The relations should be designed to satisfy the lossless join condition.
- No spurious tuples should be generated by doing a natural-join of any relations

(a)

**EMP_PROJ1**

(b)

EMP_LOCS

ENAME	PLOCATION
Smith, John B.	Bellaire
Smith, John B.	Sugarland
Narayan, Ramesh K.	Houston
English, Joyce A.	Bellaire
English, Joyce A.	Sugarland
Wong, Franklin T.	Sugarland
Wong, Franklin T.	Houston
Wong, Franklin T.	Stafford

Zelaya, Alicia J.	Stafford
Jabbar, Ahmad V.	Stafford
Wallace, Jennifer S.	Stafford
Wallace, Jennifer S.	Houston
Borg, James E.	Houston

EMP_PROJ1

SSN	PNUMBER	HOURS	PNAME	PLOCATION
123456789	1	32.5	Product X	Bellaire
123456789	2	7.5	Product Y	Sugarland
666884444	3	40.0	Product Z	Houston
453453453	1	20.0	Product X	Bellaire
453453453	2	20.0	Product Y	Sugarland
333445555	2	10.0	Product Y	Sugarland
333445555	3	10.0	Product Z	Houston
333445555	10	10.0	Computerization	Stafford
333445555	20	10.0	Reorganization	Houston

999887777	30	30.0	Newbenefits	Stafford
999887777	10	10.0	Computerization	Stafford
987987987	10	35.0	Computerization	Stafford
987987987	30	5.0	Newbenefits	Stafford
987654321	30	20.0	Newbenefits	Stafford
987654321	20	15.0	Reorganization	Houston
888665555	20	null	Reorganization	Houston

Functional Dependencies

- ◆ Functional dependencies (FDs) are used to specify *formal measures* of the "goodness" of relational designs
- ◆ FDs and keys are used to define **normal forms** for relations
- ◆ FDs are **constraints** that are derived from the *meaning* and *interrelationships* of the data attributes

Functional Dependencies (2)

- ◆ A set of attributes X *functionally determines* a set of attributes Y if the value of X determines a unique value for Y
- ◆ $X \twoheadrightarrow Y$ holds if whenever two tuples have the same value for X , they *must have* the same value for Y
If $t1[X]=t2[X]$, *then* $t1[Y]=t2[Y]$ in any relation instance $r(R)$
- ◆ $X \twoheadrightarrow Y$ in R specifies a *constraint* on all relation instances $r(R)$
- ◆ FDs are derived from the real-world constraints on the attributes

Examples of FD constraints

- ◆ Social Security Number determines employee name
 $SSN \twoheadrightarrow ENAME$
- ◆ Project Number determines project name and location
 $PNUMBER \twoheadrightarrow \{PNAME, PLOCATION\}$
- ◆ Employee SSN and project number determines the hours per week that the employee works on the project
 $\{SSN, PNUMBER\} \twoheadrightarrow HOURS$

- In EMPLOYEE relation given in Table 1,
- FD **E-ID→E-NAME** holds because for each E-ID, there is a unique value of E-NAME.
 - FD **E-ID→E-CITY** and **E-CITY→E-STATE** also holds.
 - FD **E-NAME→E-ID** **does not hold** because E-NAME 'John' is not uniquely determining E-ID.
- There are 2 E-IDs corresponding to John (E001 and E003).

EMPLOYEE

<u>E-ID</u>	E-NAME	E-CITY	E-STATE
E001	John	Delhi	Delhi
E002	Mary	Delhi	Delhi
E003	John	Noida	U.P.

- 
- ◆ The FD set for EMPLOYEE relation given in Table 1 are:

**{E-ID->E-NAME, E-ID->E-CITY,
E-ID->E-STATE, E-CITY->E-STATE}**

**X->Y will always hold if $X \supseteq Y$ it is trivial
dependency**

Eg. E-ID, E-NAME->E-ID

Else it is nontrivial FD

Functional Dependencies (3)

- ◆ An FD is a property of the attributes in the schema R
- ◆ The constraint must hold on *every relation instance* $r(R)$
- ◆ If K is a key of R , then K functionally determines all attributes in R (since we never have two distinct tuples with $t1[K]=t2[K]$)

Inference Rules for FDs

- ◆ Given a set of FDs F , we can *infer* additional FDs that hold whenever the FDs in F hold
- ◆ Armstrong's inference rules
 - A1. (Reflexive) If Y subset-of X , then $X \twoheadrightarrow Y$
 - A2. (Augmentation) If $X \twoheadrightarrow Y$, then $XZ \twoheadrightarrow YZ$
(Notation: XZ stands for $X \cup Z$)
 - A3. (Transitive) If $X \twoheadrightarrow Y$ and $Y \twoheadrightarrow Z$, then $X \twoheadrightarrow Z$
- ◆ A1, A2, A3 form a *sound* and *complete* set of inference rules

Additional Useful Inference Rules

- ◆ Decomposition
 - If $X \twoheadrightarrow YZ$, then $X \twoheadrightarrow Y$ and $X \twoheadrightarrow Z$
- ◆ Union
 - If $X \twoheadrightarrow Y$ and $X \twoheadrightarrow Z$, then $X \twoheadrightarrow YZ$
- ◆ Psuedotransitivity
 - If $X \twoheadrightarrow Y$ and $WY \twoheadrightarrow Z$, then $WX \twoheadrightarrow Z$
- ◆ **Closure** of a set F of FDs is the set F^+ of all FDs that can be inferred from F

Q. Given FD set of a Relation R, The attribute closure set S be the set of A

- ◆ Add A to S.
- ◆ Recursively add attributes which can be functionally determined from attributes of the set S until done.

- ◆ From Table 1, FDs are

Given R(E-ID, E-NAME, E-CITY, E-STATE)

FDs = { E-ID->E-NAME, E-ID->E-CITY, E-ID->E-STATE, E-CITY->E-STATE }

The attribute closure of E-ID can be calculated as:

- ◆ Add E-ID to the set {E-ID}
- ◆ Add Attributes which can be derived from any attribute of set. In this case, E-NAME and E-CITY, E-STATE can be derived from E-ID. So these are also a part of closure.
- ◆ As there is one other attribute remaining in relation to be derived from E-ID. So result is:
- ◆ $(E-ID)^+ = \{E-ID, E-NAME, E-CITY, E-STATE\}$ Similarly,
- ◆ $(E-NAME)^+ = \{E-NAME\}$ $(E-CITY)^+ = \{E-CITY, E_STATE\}$

Find the attribute closures of given FDs

$R(ABCDE) = \{AB \rightarrow C, B \rightarrow D, C \rightarrow E, D \rightarrow A\}$

To find $(B)^+$, we will add attribute in set using various FD which has been shown in table below.

Attributes Added in Closure	FD used
$\{B\}$	Triviality
$\{B, D\}$	$B \rightarrow D$
$\{B, D, A\}$	$D \rightarrow A$
$\{B, D, A, C\}$	$AB \rightarrow C$
$\{B, D, A, C, E\}$	$C \rightarrow E$

- We can find $(C, D)^+$ by adding C and D into the set (triviality) and then E using $(C \rightarrow E)$ and then A using $(D \rightarrow A)$ and set becomes. $(C, D)^+ = \{C, D, E, A\}$
- Similarly we can find $(B, C)^+$ by adding B and C into the set (triviality) and then D using $(B \rightarrow D)$ and then E using $(C \rightarrow E)$ and then A using $(D \rightarrow A)$ and set becomes $(B, C)^+ = \{B, C, D, E, A\}$

The attribute(s) whose closure constitutes all the attributes in the given relation are candidate keys

- ◆ $R = (A, B, C, D, E, H)$ on which the following functional dependencies hold: $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$. What are the candidate keys of R ? [GATE 2005]

- (a) AE, BE
- (b) AE, BE, DE
- (c) AEH, BEH, BCH
- (d) AEH, BEH, DEH

- ◆ **Answer:** $(AE)^+ = \{ABECD\}$ which is not set of all attributes. So AE is not a candidate key. Hence option A and B are wrong.
 $(AEH)^+ = \{ABCDEH\}$
 $(BEH)^+ = \{BEHCDA\}$
 $(BCH)^+ = \{BCHDA\}$ which is not set of all attributes. So BCH is not a candidate key. Hence option C is wrong.
So correct answer is D.

3.1 Normalization of Relations (1)

- **Normalization:**

- The process of decomposing unsatisfactory "**bad**" relations by breaking up their attributes into smaller relations
 - Resulting designs are of high quality

- **Normal form:**

- Condition using keys and FDs of a relation to certify whether a relation schema is in a particular normal form

- **De-normalization:**

- The process of storing the join of higher normal form relations as a base relation—which is in a lower normal form
 - For Performance Enhancement

Introduction to Normalization

- ◆ **Normalization:** Process of decomposing unsatisfactory "bad" relations by breaking up their attributes into smaller relations
- ◆ **Normal form:** Condition using keys and FDs of a relation to certify whether a relation schema is in a particular normal form
 - 2NF, 3NF, BCNF based on keys and FDs of a relation schema
 - 4NF based on keys, multi-valued dependencies

First Normal Form

- ◆ Disallows composite attributes, multivalued attributes, and **nested relations**; attributes whose values *for an individual tuple* are non-atomic
- ◆ Considered to be part of the definition of relation

(a)

DEPARTMENT

DNAME	<u>DNUMBER</u>	DMGRSSN	DLOCATIONS
			

(b)

DEPARTMENT

DNAME	<u>DNUMBER</u>	DMGRSSN	DLOCATIONS
Research	5	333445555	{Bellaire, Sugarland, Houston}
Administration	4	987654321	{Stafford}
Headquarters	1	888665555	{Houston}

(c)

DEPARTMENT

DNAME	<u>DNUMBER</u>	DMGRSSN	<u>DLOCATION</u>
Research	5	333445555	Bellaire
Research	5	333445555	Sugarland
Research	5	333445555	Houston
Administration	4	987654321	Stafford
Headquarters	1	888665555	Houston

(a) **EMP_PROJ**

SSN	ENAME	PROJS	
		PNUMBER	HOURS

(b) **EMP_PROJ**

SSN	ENAME	PNUMBER	HOURS
123456789	Smith,John B.	1	32.5
		2	7.5
666884444	Narayan,Ramesh K.	3	40.0
453453453	English,Joyce A.	1	20.0
		2	20.0
333445555	Wong,Franklin T.	2	10.0
		3	10.0
		10	10.0
		20	10.0
999887777	Zelaya,Alicia J.	30	30.0
		10	10.0
987987987	Jabbar,Ahmad V.	10	35.0
		30	5.0
987654321	Wallace,Jennifer S.	30	20.0
		20	15.0
888665555	Borg,James E.	20	null

(c) **EMP_PROJ1**

<u>SSN</u>	ENAME
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EMP_PROJ2

<u>SSN</u>	<u>PNUMBER</u>	HOURS
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Second Normal Form

- ◆ Uses the concepts of FDs, primary key
- ◆ Definitions:
 - **Prime attribute** - attribute that is member of the primary key K
 - **Full functional dependency** - a FD $Y \twoheadrightarrow Z$ where removal of any attribute from Y means the FD does not hold any more

(a)

EMP_PROJ

<u>SSN</u>	ENAME	PROJS	
		<u>PNUMBER</u>	HOURS

- ◆ {SSN, PNUMBER} is key of relation
- ◆ Check the closure of {SSN,PUMBER}
- ◆ Set of functional dependencies is given as

{SSN, PNUMBER} $\twoheadrightarrow$ HOURS

{SSN} $\twoheadrightarrow$ Ename

Examples

Second Normal Form

- ◆ $\{SSN, PNUMBER\} \twoheadrightarrow HOURS$ is a full FD since neither $SSN \twoheadrightarrow HOURS$ nor $PNUMBER \twoheadrightarrow HOURS$ hold
- ◆ $\{SSN, PNUMBER\} \twoheadrightarrow ENAME$ is *not* a full FD (it is called a *partial dependency*) since $SSN \twoheadrightarrow ENAME$ also holds
- ◆ A relation schema R is in **second normal form (2NF)** if every non-prime attribute A in R is fully functionally dependent on the primary key
- ◆ R can be decomposed into 2NF relations via the process of 2NF normalization

A relation that is not in 2NF

ACTIVITY

<u>Student_ID</u>	<u>Activity</u>	Fee

Key: Student_ID, Activity

Activity → Fee

Fee is determined by Activity

Student_ID	Activity	Fee
222-22-2020	Swimming	30
232-22-2111	Golf	100
222-22-2020	Golf	100
255-24-2332	Hiking	50

2NF decomposition

Student_ID

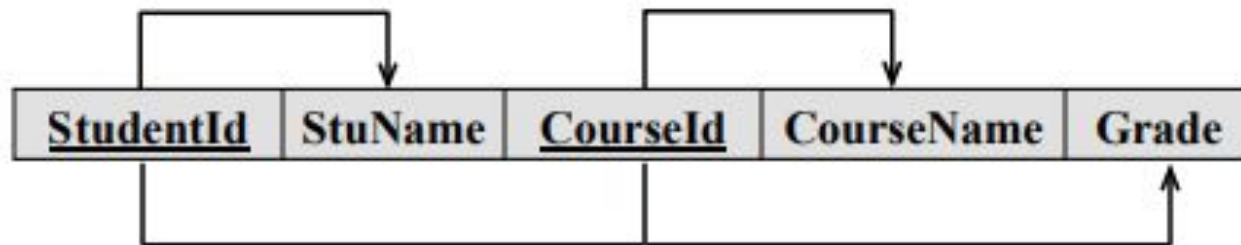
Activity

Activity

Fee

Examples of 2NF decomposition

Functional Dependencies in Student



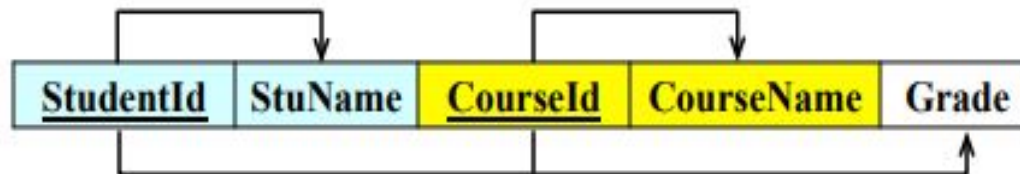
Can represent FDs with arrows as above, or

- $\text{StudentId} \rightarrow \text{StuName}$,
- $\text{CourseId} \rightarrow \text{CourseName}$
- $\text{StudentId}, \text{CourseId} \rightarrow \text{Grade}$ (and $\text{StuName}, \text{CourseName}$)

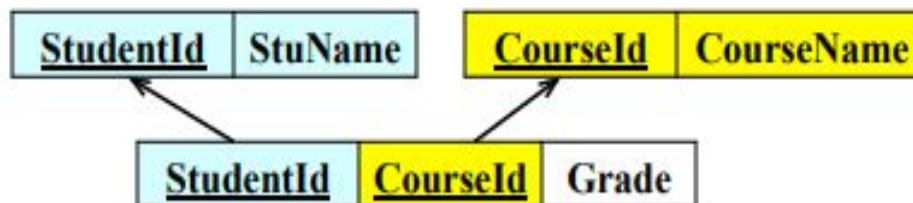
Therefore, NOT in 2nd Normal Form!!

2NF: Normalizing

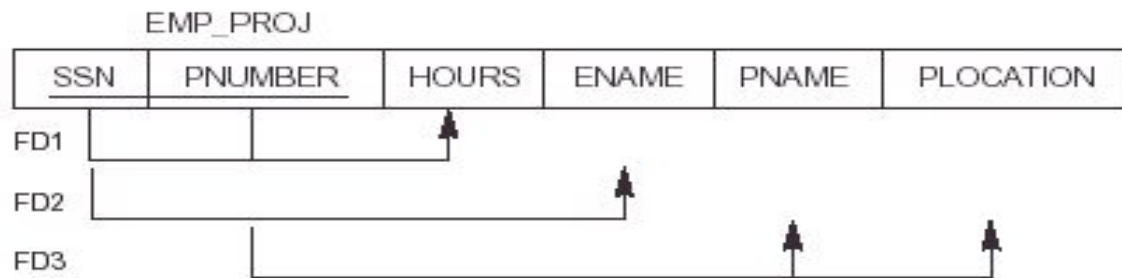
- How do we convert the partial dependencies into normal ones ? By breaking into more tables.



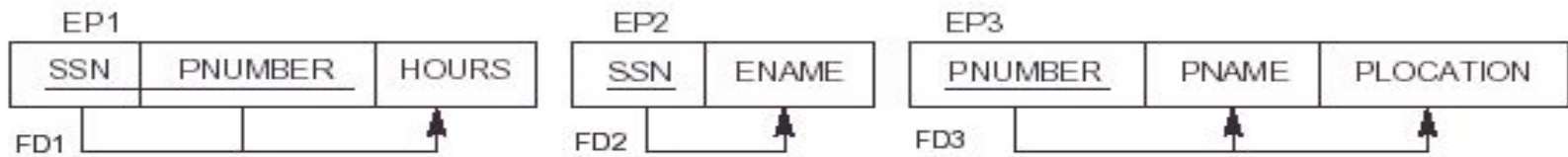
- Becomes ... (notice above arrows mean functional dependency, below they mean FK constraints)



(a)



2NF NORMALIZATION



Third Normal Form

◆ Definition

- **Transitive functional dependency** – if there a set of attribute Z that are neither a primary or candidate key (are non prime) and both $X \twoheadrightarrow Z$ and $Y \twoheadrightarrow Z$ holds.

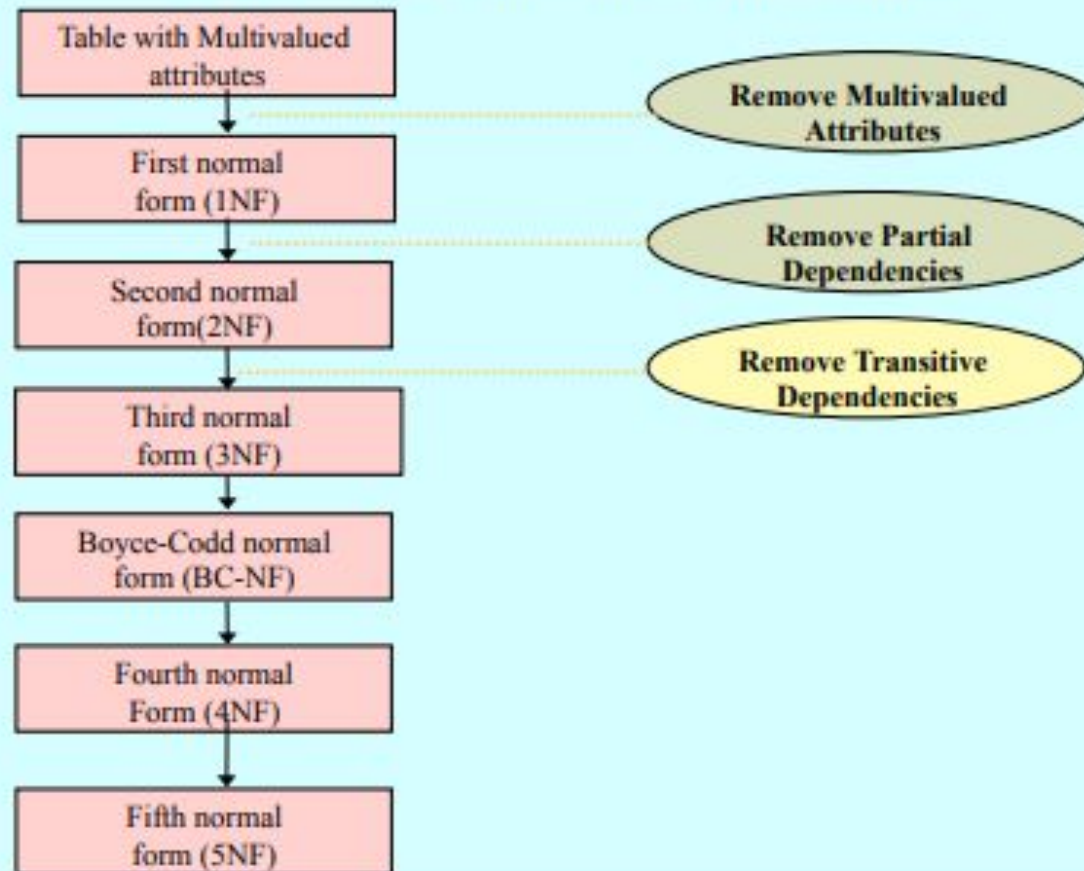
◆ Examples:

- $SSN \twoheadrightarrow DMGRSSN$ is a transitive FD since $SSN \twoheadrightarrow DNUMBER$ and $DNUMBER \twoheadrightarrow DMGRSSN$ hold
- $SSN \twoheadrightarrow ENAME$ is *non-transitive* since there is no set of attributes X where $SSN \twoheadrightarrow X$ and $X \twoheadrightarrow ENAME$

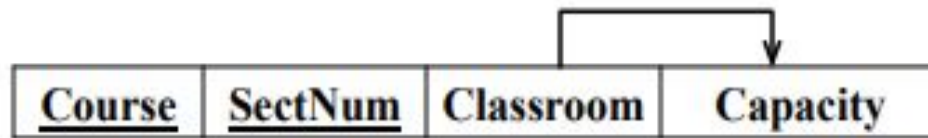
3rd Normal Form

A relation schema R is in **third normal form (3NF)** if it is in 2NF *and* no non-prime attribute A in R is transitively dependent on the primary key

Figure: 4-22 Steps in Normalization



3NF Example

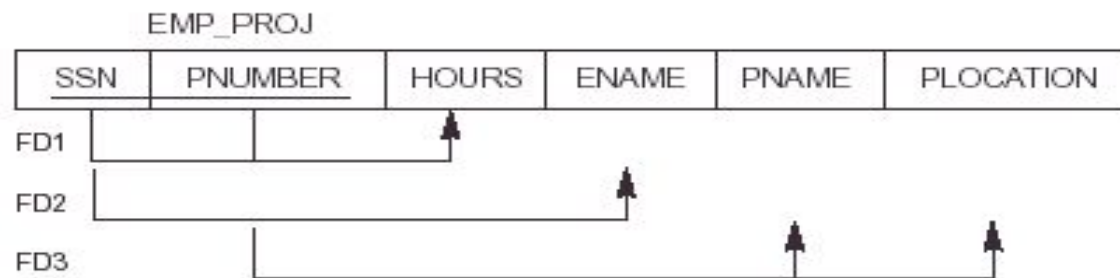


(Course, SectNum) $\square$ Classroom

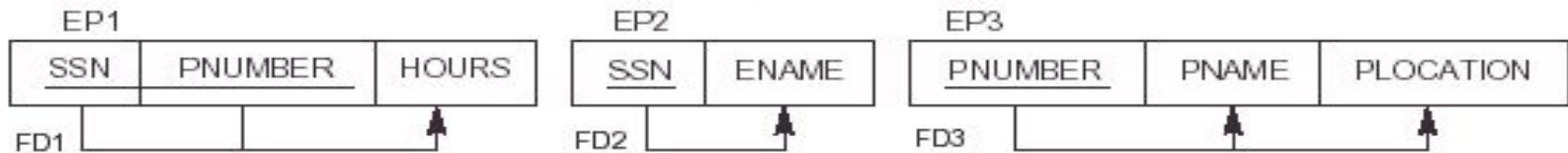
- Classroom $\rightarrow$ Capacity TRANSITIVE
- Any partial FDs? NO
- Any transitive FDs? YES !
 - How do we eliminate it?
 - By breaking into its own table

<u>Course</u>	<u>SectNum</u>	Classroom
<u>Classroom</u>	Capacity	

(a)



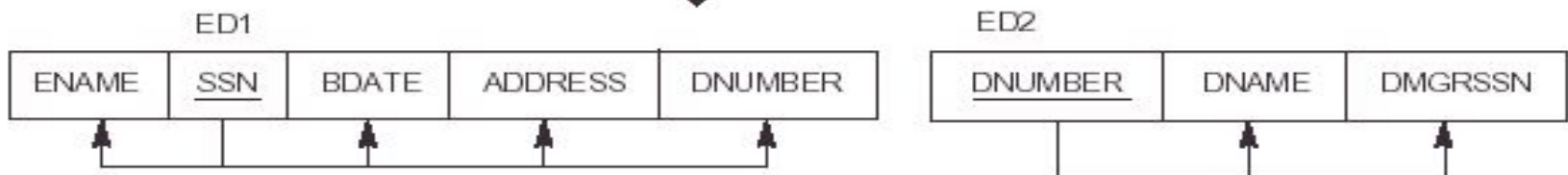
2NF NORMALIZATION



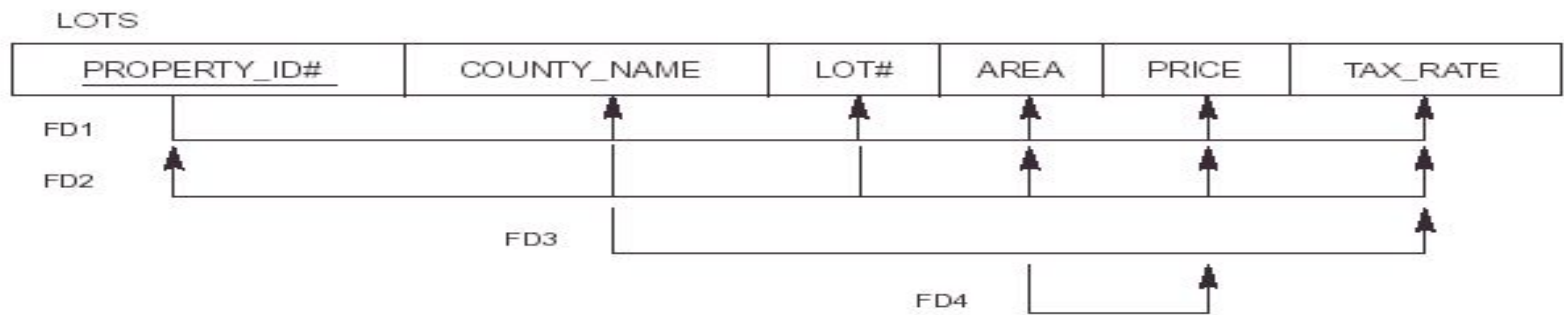
(b)



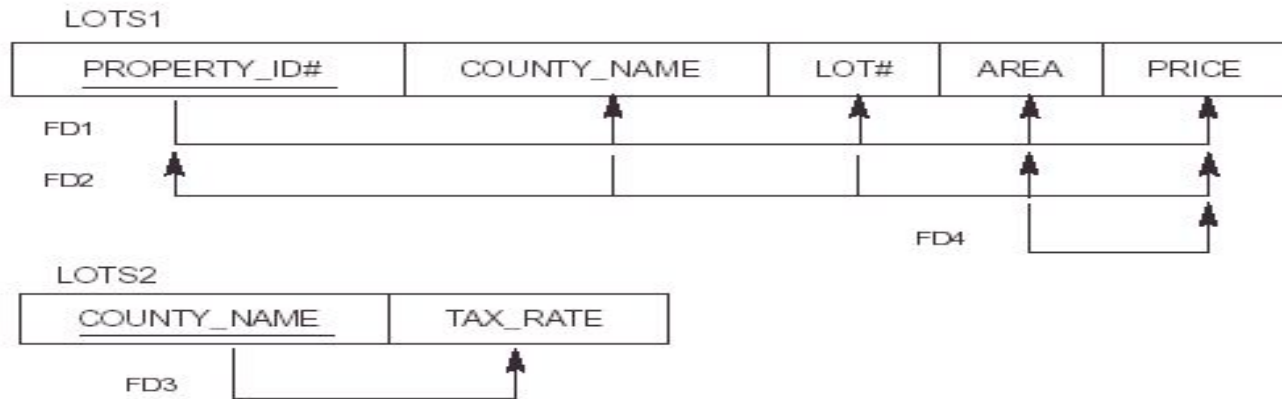
3NF NORMALIZATION



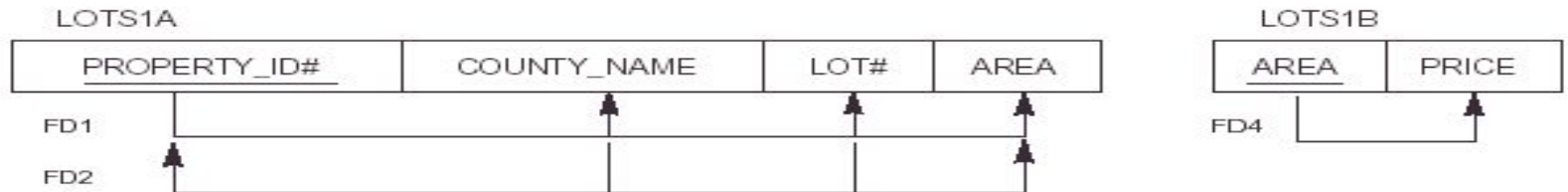
(a)



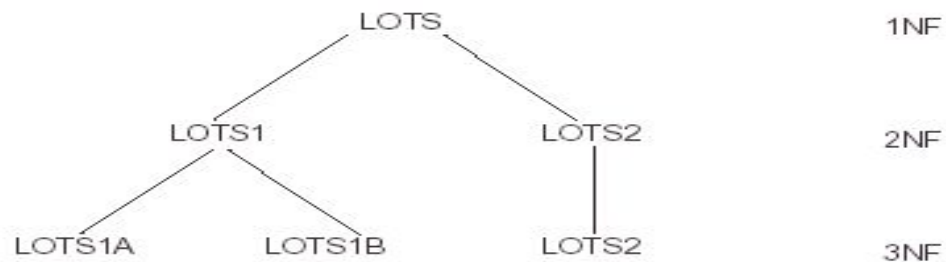
(b)



(c)



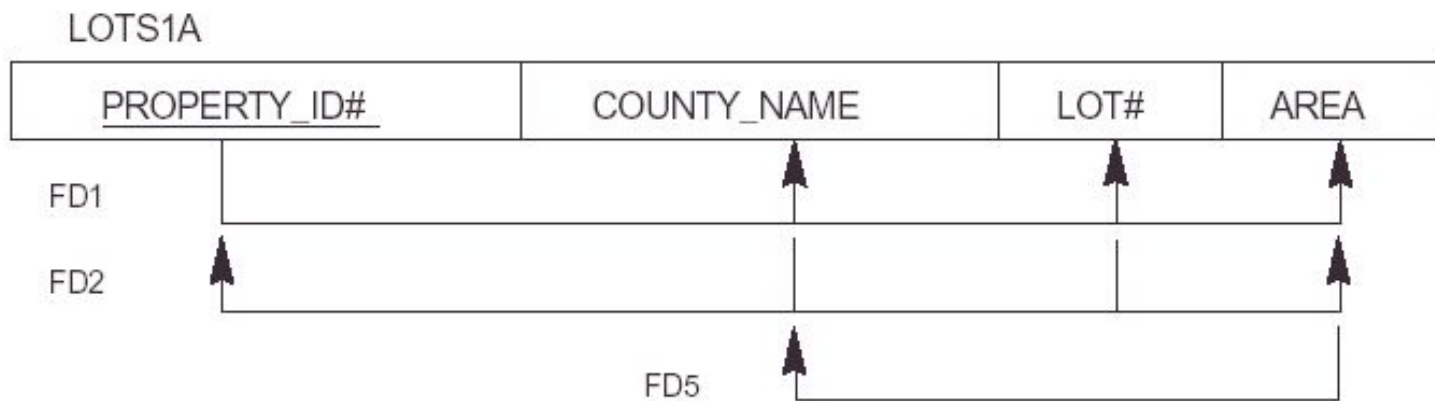
(d)



BCNF (Boyce-Codd Normal Form)

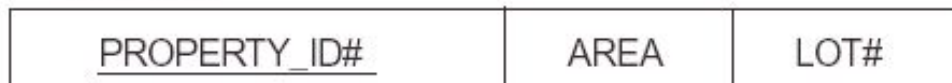
- ◆ A relation schema R is in **Boyce-Codd Normal Form (BCNF)** if whenever an FD $X \twoheadrightarrow A$ holds in R , then X is a superkey of R
 - Each normal form is strictly stronger than the previous one:
 - Every 2NF relation is in 1NF
 - Every 3NF relation is in 2NF
 - Every BCNF relation is in 3NF
 - There exist relations that are in 3NF but not in BCNF
 - The goal is to have each relation in BCNF (or 3NF)

(a)



BCNF Normalization

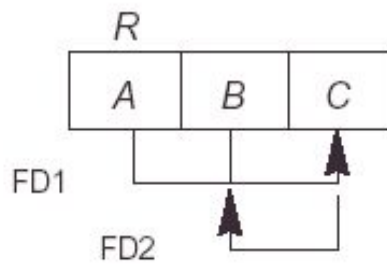
LOTS1AX



LOTS1AY



(b)



(A,C) (C,B)
 (A,B)

BCNF

- Boyce-Codd normal form (BCNF)

A relation is in BCNF, if and only if,
every determinant is a candidate key.

- The difference between 3NF and BCNF is that
for a functional dependency $A \twoheadrightarrow B$, 3NF allows this dependency
in a relation if

B is a primary-key attribute and A is not a candidate key,
whereas

BCNF insists that for this dependency to remain in a relation, A
must be a candidate key

TEACH

STUDENT	COURSE	INSTRUCTOR
Narayan	Database	Mark
Smith	Database	Navathe
Smith	Operating Systems	Ammar
Smith	Theory	Schulman
Wallace	Database	Mark
Wallace	Operating Systems	Ahamad
Wong	Database	Omiecinski
Zelaya	Database	Navathe

BCNF

- ◆ {Student,course} □ Instructor
- ◆ Instructor □ Course
- ◆ Decomposing into 2 schemas
 - {Student,Instructor} {Student,Course}
 - {Course,Instructor} {Student,Course}
 - {Course,Instructor} {Instructor,Student}

TABLE 4-8 Shipment Relation

Shipment#	Origin	Destination	Distance
409	Seattle	Denver	1,537
618	Chicago	Dallas	1,058
723	Boston	Atlanta	1,214
824	Denver	Los Angeles	975
629	Seattle	Denver	1,537

Insertion anomaly?
Deletion anomaly?
Modification anomaly?

1. Develop a diagram that shows the functional dependencies in the SHIPMENT relation.
2. In what normal form is SHIPMENT? Why?
3. Convert SHIPMENT to 3NF if necessary. Show the resulting table(s) with the sample data presented in SHIPMENT.

4NF

- **A multi-valued dependency exists when**
 - There are at least 3 attributes A, B, C in a relation and
 - For each value of A there is a well defined set of values for B, and a well defined set of values for C,
 - But the set of values for B is independent on the set of values for C
- **4NF = 3NF with no multi-valued dependency**

4NF Example

- Assume that
 - Each subject is taught by many Instructors
 - The same books are used in many subjects
 - Each Instructor uses a different book

Course, Instructor $\rightarrow$ Text

Course, Text $\rightarrow$ Instructor

Course	Instructor	Text
CS 121	Dr. A. James	Int to Com Science
CS 121	Dr. P. Hold	Comp Scien Int

Course	Instructor	Text
CS 141	Dr. T. Watson	Int to Com Science
CS 141	Dr. P. Hold	Comp Scien Int
CS 101	Dr. M. Jones	COMP SCIEN

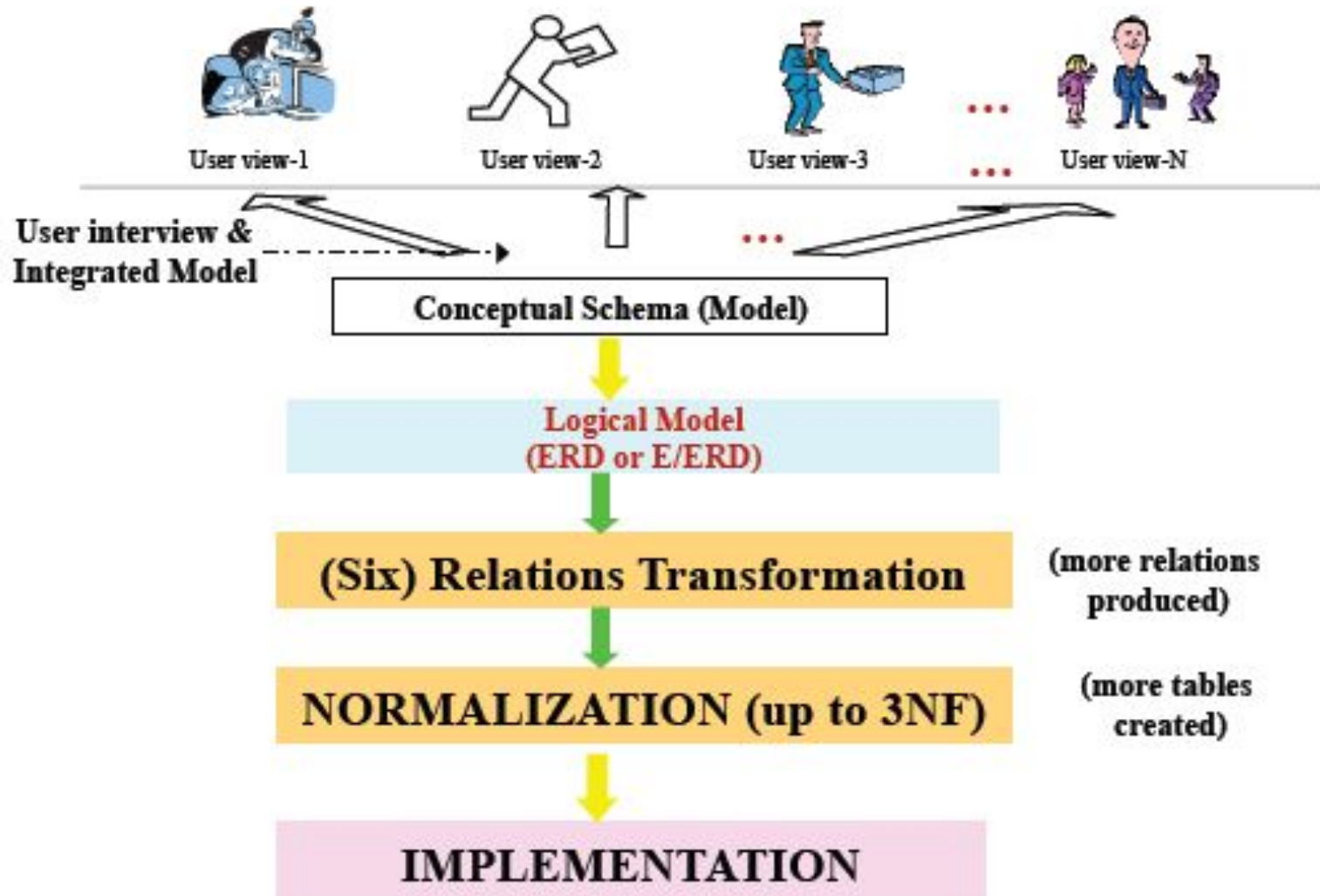
4NF: Decompose into (Course, Instructor) and (Course, Text)

Logical Database Design

You have just learned and completed one of the most important concepts and theories, **integrity constraints** and **normalization**, for developing a quality of database.



Steps of Database Development



The Normalization Example in the Text Book

Figure 4-24 INVOICE (Pine Valley Furniture Company)

PVFC Customer Invoice

Customer ID	2	Order ID	1006
Customer Name	Value Furniture	Order Date	10/24/2010
Address	15145 S.W. 17th St. Plano TX 75022		

Product ID	Product Description	Finish	Quantity	Unit Price	Extended Price
7	Dining Table	Natural Ash	2	\$800.00	\$1,600.00
5	Writer's Desk	Cherry	2	\$325.00	\$650.00
4	Entertainment Center	Natural Maple	1	\$650.00	\$650.00
				Total	\$2,900.00

Figure 4-25 INVOICE Data

Table with multivalued attributes, not in 1st normal form

<u>OrderID</u>	Order Date	Customer ID	Customer Name	Customer Address	<u>ProductID</u>	Product Description	Product Finish	Product StandardPrice	Ordered Quantity
1006	10/24/2010	2	Value Furniture	Plano, TX	7	Dining Table	Natural Ash	800.00	2
					5	Writer's Desk	Cherry	325.00	2
					4	Entertainment Center	Natural Maple	650.00	1
1007	10/25/2010	6	Furniture Gallery	Boulder, CO	11	4-Dr Dresser	Oak	500.00	4
					4	Entertainment Center	Natural Maple	650.00	3

Note: this is NOT a relation. WHY?

Figure 4-26 INVOICE relation (1NF)

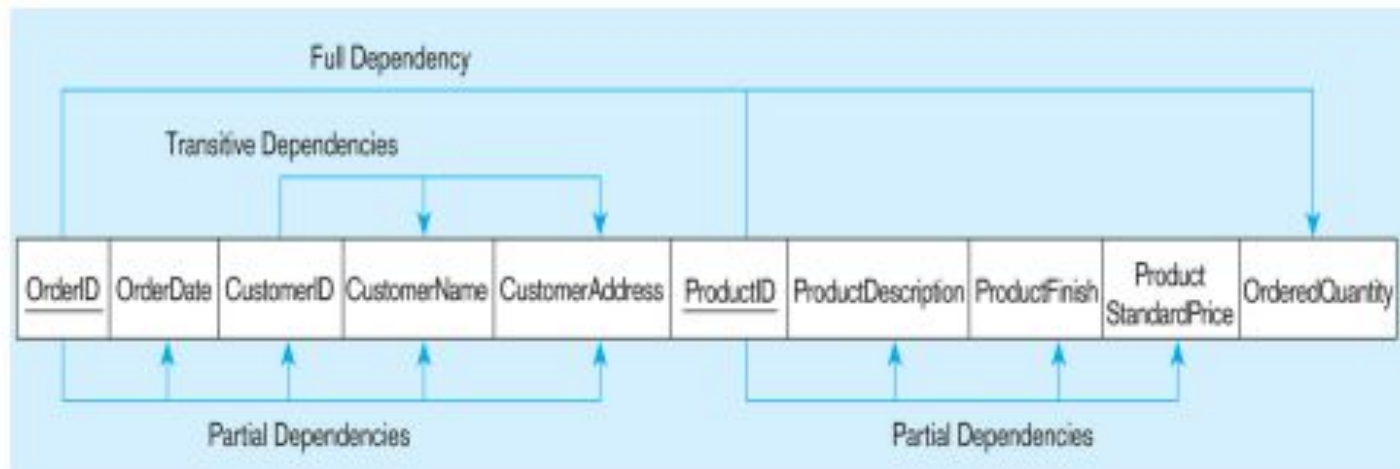
Table with no multivalued attributes and unique rows

<u>OrderID</u>	Order Date	Customer ID	Customer Name	Customer Address	<u>ProductID</u>	Product Description	Product Finish	Product StandardPrice	Ordered Quantity
1006	10/24/2010	2	Value Furniture	Plano, TX	7	Dining Table	Natural Ash	800.00	2
1006	10/24/2010	2	Value Furniture	Plano, TX	5	Writer's Desk	Cherry	325.00	2
1006	10/24/2010	2	Value Furniture	Plano, TX	4	Entertainment Center	Natural Maple	650.00	1
1007	10/25/2010	6	Furniture Gallery	Boulder, CO	11	4-Dr Dresser	Oak	500.00	4
1007	10/25/2010	6	Furniture Gallery	Boulder, CO	4	Entertainment Center	Natural Maple	650.00	3

Note: this is relation, but not a well-structured one. WHY?

Anomalies in this Table

- **Insertion**—if new product is ordered for order 1007 of existing customer, customer data must be re-entered, causing duplication
- **Deletion**—if we delete the Dining Table from Order 1006, we lose information concerning this item's finish and price
- **Update**—changing the price of product ID 4 requires update in several records



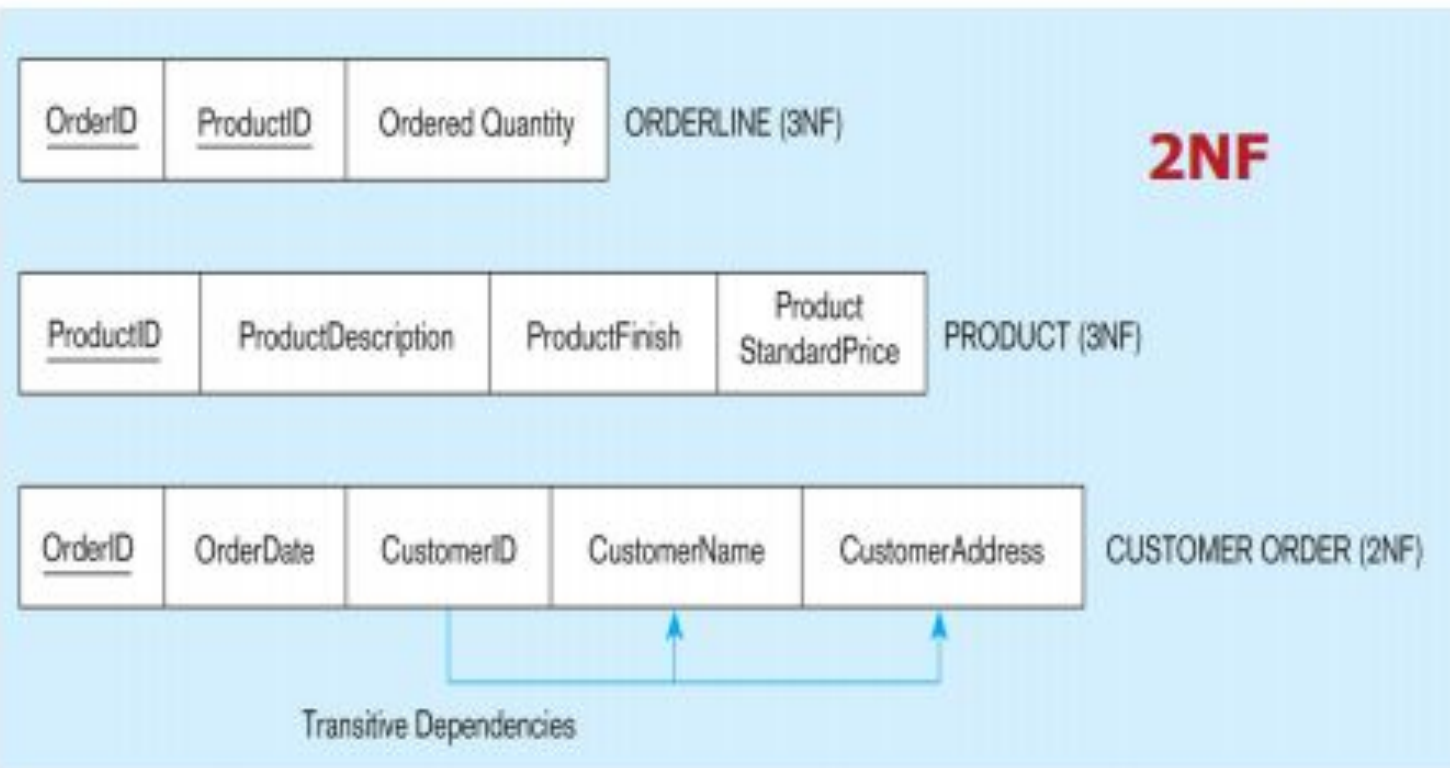
Order_ID ➔ Order_Date, Customer_ID, Customer_Name, Customer_Address

Customer_ID ➔ Customer_Name, Customer_Address

Product_ID ➔ Product_Description, Product_Finish, Unit_Price

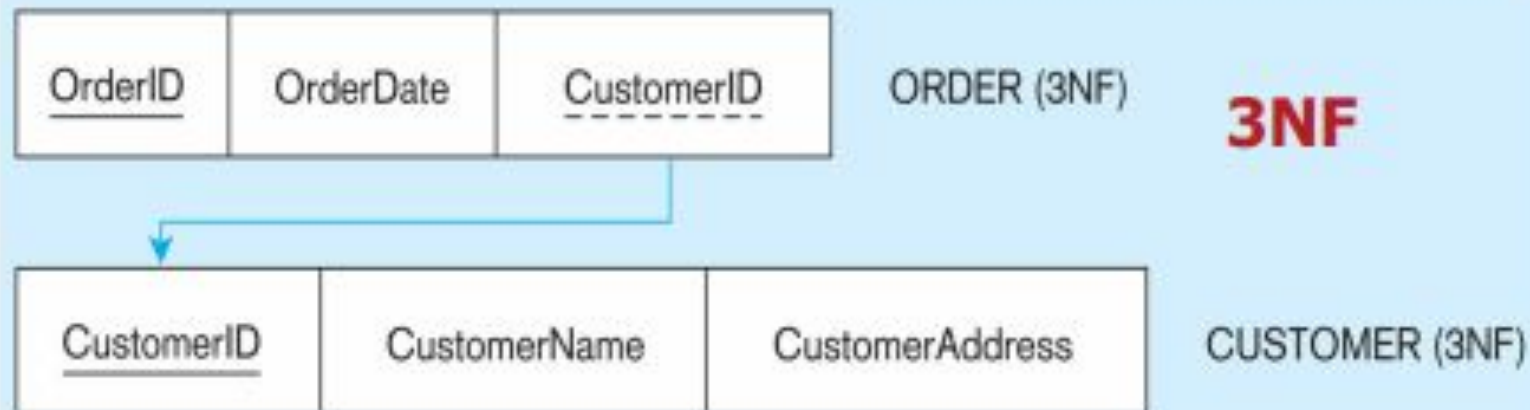
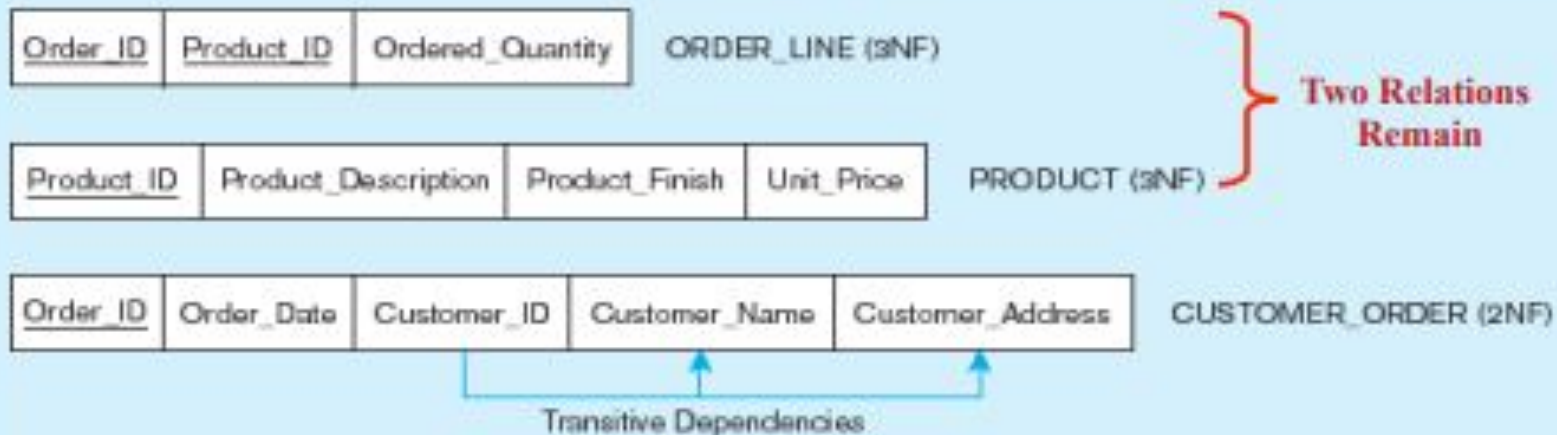
Order_ID, Product_ID ➔ Order_Quantity

Figure 4-28 Partial Dependencies were Removed (2NF)



Partial dependencies are removed, but there are still transitive dependencies

Figure 4-29 Transitive Dependencies were Removed (3NF)



Example

- ◆ Given the relation

Book(Book_title, Authorname, Book_type,
Listprice, Author_affil, Publisher)

The FDs are

Book_title $\square$ Publisher, Book_type

Book_type $\square$ Listprice

Authorname $\square$ Author_affil

(Book_title, Publisher, Book_type)

(Book_type, Listprice)

(Authorname, Author_affil)

(Book_title, Authorname)



Example



- ◆ What normal form the relation in?

- 
- 2 NF does not allow partial dependency.
 - 3NF does not allow transitive dependency.
 - BCNF does not allow anything other than super key as determinant.

2NF

- ◆ Not allowed FDs –
 - ◆ prime $\rightarrow$ non prime
 - ◆ If say your functional dependency is of the form $A \rightarrow X$ where 'A' is a prime attribute but not a key and 'X' is a non prime attribute, then such an FD is not allowed in 2NF.
- ◆ Allowed FDs –
 - ◆ Prime $\rightarrow$ Prime
 - ◆ Non prime $\rightarrow$ Prime/Non prime
 - ◆ Key $\rightarrow$ Prime/Non prime
 - ◆ Prime + Non prime combination $\rightarrow$ Prime/Non Prime

2. Third Normal Form (3NF) :

- ◆ Not allowed FDs –
 - ◆ Prime $\rightarrow$ Non prime (2NF requirement)
 - ◆ Non prime $\rightarrow$ Non prime (3NF special)
- ◆ Allowed FDs –
 - ◆ Prime $\rightarrow$ Prime
 - ◆ Non prime $\rightarrow$ Prime
 - ◆ Key $\rightarrow$ Prime/Non prime
 - ◆ Prime + Non Prime $\rightarrow$ Prime/Non prime

3. Boyce-Codd Normal Form (BCNF) :

- ◆ Not allowed FDs –
 - ◆ Prime $\rightarrow$ Non prime
 - ◆ Non prime $\rightarrow$ Non prime
 - ◆ Prime $\rightarrow$ Prime
 - ◆ Non prime $\rightarrow$ Prime
- ◆ Allowed –
 - ◆ Key $\rightarrow$ Prime/Non prime